



Smart Attendance System

Using ESP32 and Fingerprint sensor

Developed By

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Smart Attendance System Using Fingerprint

1. Team Members

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2. Project Overview

The Smart Attendance System is designed to automate attendance using fingerprint recognition technology.

It improves accuracy, security, and efficiency compared to manual methods. The system scans fingerprints using a fingerprint sensor. If matched with stored data, attendance is recorded automatically.

All data is stored in real-time using Google Sheets for easy access and tracking.

Main Features:

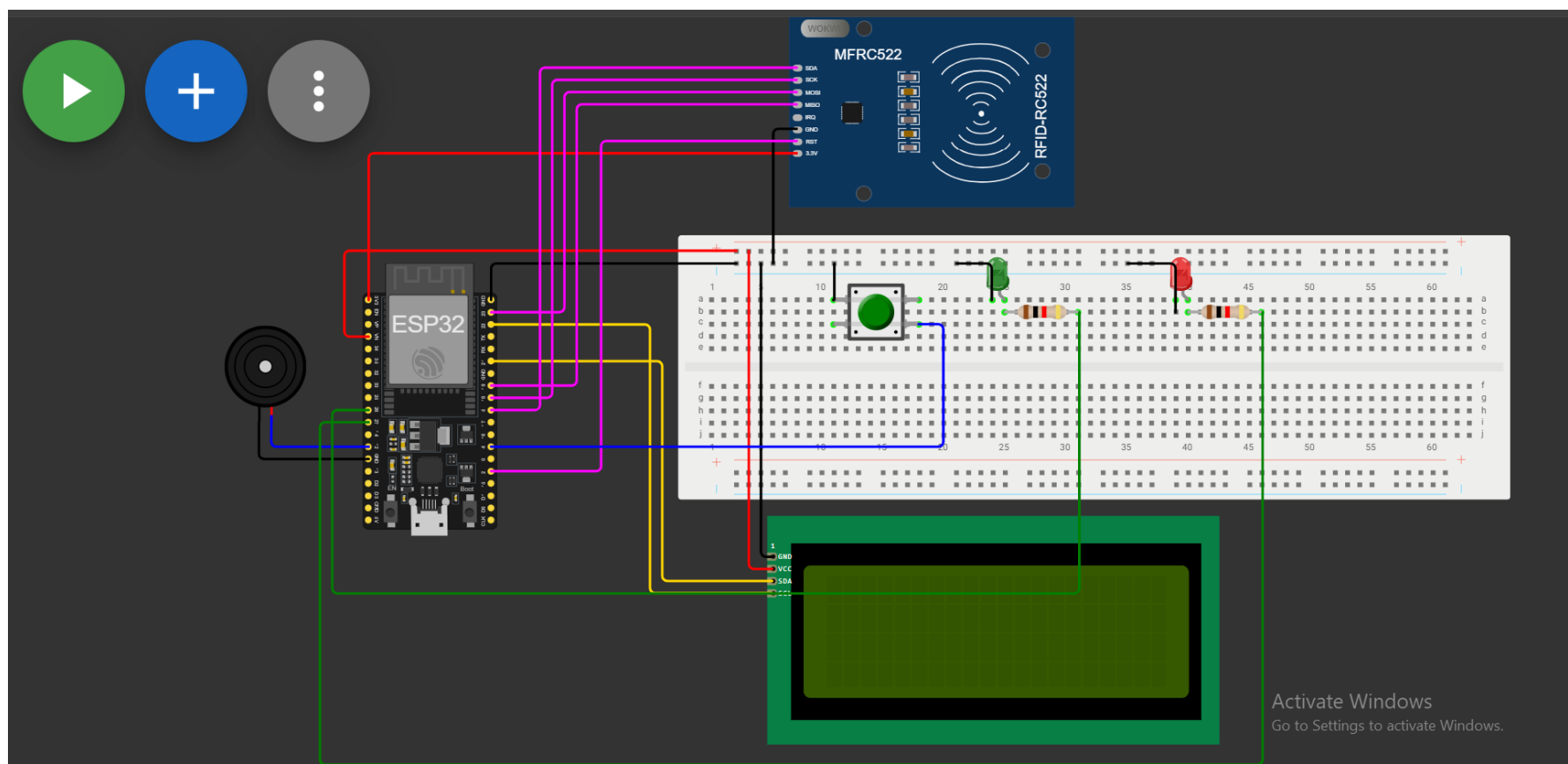
- Fast automated attendance
- High security fingerprint verification
- Real-time cloud storage via Google Sheets
- Reduces human error and fraud

3. Circuit

Components Used:

- ESP32
- Fingerprint Sensor Module
- Buzzer
- LCD Display
- Button
- Breadboard

Circuit Diagram:





Description:

- **Microcontroller:** ESP32 (Handles WiFi and Logic).
- **Biometrics:** Adafruit Fingerprint Sensor (Optical scanning).
- **Display:** LCD 16x2 with I2C (User interface).
- **Cloud Integration:** Google Apps Script (Web App URL).
- **Feedback:** Active Buzzer for different sound alerts.

4. Code

```
sketch_may11a.ino
1  #include <WiFi.h>
2  #include <HttpClient.h>
3  #include <Wire.h>
4  #include <LiquidCrystal_I2C.h>
5  #include <Adafruit_Fingerprint.h>
6
7  // ===== WIFI & GOOGLE SHEET =====
8  const char* ssid = "3bouda";
9  const char* password = "#Boody2005";
10
11  String googleScriptURL =
12  "https://script.google.com/macros/s/AKfycbwefDKS80oP5ktOnrM3dg7kCR91NbMKUNsg7HU8BIR6-wEbuPNWoB2cXL1Az3qU8ak9/exec";
13
14  // ===== PINS =====
15  #define RXD2 16
16  #define TXD2 17
17
18  #define BUTTON_PIN 4
19  #define BUZZER_PIN 12
20
21  // LEDs
22  #define GREEN_LED 26
23  #define RED_LED 27
24
25  // ===== LCD & FINGERPRINT =====
26  LiquidCrystal_I2C lcd(0x27, 16, 2);
27  Adafruit_Fingerprint finger = Adafruit_Fingerprint(&Serial2);
28
29  // =====
30  // ===== LED FUNCTIONS =====
31  // =====
32
33  void greenSuccess() {
34    digitalWrite(GREEN_LED, HIGH);
35    delay(500);
36    digitalWrite(GREEN_LED, LOW);
37  }
```



sketch_may11a.ino

```
39 void redFail() {
40     digitalWrite(RED_LED, HIGH);
41     delay(500);
42     digitalWrite(RED_LED, LOW);
43 }
44
45 // =====
46 // ===== SOUND FUNCTIONS =====
47 // =====
48
49 void soundSuccess() {
50     tone(BUZZER_PIN, 2000, 200);
51     greenSuccess();
52 }
53
54 void soundFail() {
55     tone(BUZZER_PIN, 400, 150);
56     delay(100);
57     tone(BUZZER_PIN, 400, 150);
58     redFail();
59 }
60
61 void soundAlert() {
62     tone(BUZZER_PIN, 1200, 100);
63 }
64
65 void soundDelete() {
66     for (int i = 0; i < 3; i++) {
67         tone(BUZZER_PIN, 800, 100);
68         delay(150);
69     }
70 }
71
72 void soundSaveMelody() {
73     tone(BUZZER_PIN, 1500, 100);
74     delay(100);
```



sketch_may11a.ino

```
75
76   tone(BUZZER_PIN, 2000, 100);
77   delay(100);
78
79   tone(BUZZER_PIN, 2500, 300);
80
81   greenSuccess();
82 }
83
84 // =====
85 // ===== SETUP =====
86 // =====
87
88 void setup() {
89
90   Serial.begin(115200);
91
92   Serial2.begin(57600, SERIAL_8N1, RXD2, TXD2);
93
94   pinMode(BUTTON_PIN, INPUT_PULLUP);
95   pinMode(BUZZER_PIN, OUTPUT);
96
97   pinMode(GREEN_LED, OUTPUT);
98   pinMode(RED_LED, OUTPUT);
99
100   digitalWrite(GREEN_LED, LOW);
101   digitalWrite(RED_LED, LOW);
102
103   lcd.init();
104   lcd.backlight();
105
106   // ===== WIFI CONNECT =====
107
108   lcd.setCursor(0, 0);
109   lcd.print("WiFi Connecting");
110
111   WiFi.begin(ssid, password);
```

sketch_may11a.ino

```
113   while (WiFi.status() != WL_CONNECTED) {
114
115       delay(500);
116
117       lcd.setCursor(0, 1);
118       lcd.print("Please Wait...");
119   }
120
121   lcd.clear();
122   lcd.print("WiFi Connected");
123   soundSuccess();
124
125   // ===== SENSOR CHECK =====
126
127   if (finger.verifyPassword()) {
128
129       lcd.setCursor(0, 1);
130       lcd.print("Sensor Ready");
131
132       soundSuccess();
133   } else {
134
135       lcd.clear();
136       lcd.print("Sensor Error");
137
138       while (1) {
139
140           tone(BUZZER_PIN, 200, 500);
141
142           digitalWrite(RED_LED, HIGH);
143           delay(500);
144
145           digitalWrite(RED_LED, LOW);
146           delay(500);
147       }
148   }
149 }
```

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```
151   delay(1500);
152
153   lcd.clear();
154 }
155
156 // =====
157 // ===== LOOP =====
158 // =====
159
160 void loop() {
161
162   lcd.setCursor(0, 0);
163   lcd.print("Scan Finger...");
164
165   lcd.setCursor(0, 1);
166   lcd.print("Ready To Work ");
167
168   // ===== ENROLL NEW USER =====
169
170   if (digitalRead(BUTTON_PIN) == LOW) {
171
172       soundAlert();
173
174       delay(500);
175
176       finger.getTemplateCount();
177
178       enrollFinger(finger.templateCount + 1);
179   }
180
181   // ===== READ FINGER =====
182
183   int fingerID = getFingerprintID();
184
185   if (fingerID >= 0) {
186
187       processAttendance(fingerID);
```

sketch_may11a.ino

```
191 // =====
192 // ===== PROCESS ATTENDANCE =====
193 // =====
194
195 void processAttendance(int id) {
196
197   if (WiFi.status() == WL_CONNECTED) {
198
199       HTTPClient http;
200
201       String url =
202         googleScriptURL +
203         "?action=login&id=" +
204         String(id);
205
206       lcd.clear();
207       lcd.print("Syncing Cloud");
208
209       http.begin(url);
210
211       http.setFollowRedirects(HTTPC_STRICT_FOLLOW_REDIRECTS);
212
213       int httpCode = http.GET();
214
215       // ===== SUCCESS HTTP =====
216
217       if (httpCode > 0) {
218
219           String response = http.getString();
220
221           Serial.println(response);
222
223           // ===== DELETE COMMAND =====
224
225           if (response == "COMMAND:DELETE") {
226
227               if (finger.deleteModel(id) == FINGERPRINT_OK) {
```




```
sketch_may11a.ino

229     lcd.clear();
230
231     lcd.print("ID ");
232     lcd.print(id);
233
234     lcd.setCursor(0, 1);
235     lcd.print("Deleted");
236
237     soundDelete();
238 }
239 }
240
241 // ===== DUPLICATE ATTENDANCE =====
242
243 else if (response.startsWith("Already In")) {
244
245     int split = response.indexOf('|');
246
247     String name = response.substring(split + 1);
248
249     lcd.clear();
250
251     lcd.print("Wait 5 Mins");
252
253     lcd.setCursor(0, 1);
254     lcd.print(name);
255
256     soundFail();
257 }
258
259 // ===== SUCCESS =====
260
261 else {
262
263     int split = response.indexOf('|');
264
265     String msg = response.substring(0, split);
```

```
sketch_may11a.ino

265     String msg = response.substring(0, split);
266     String grp = response.substring(split + 1);
267     lcd.clear();
268     lcd.print(msg);
269     lcd.setCursor(0, 1);
270     lcd.print(grp);
271     soundSuccess();
272 }
273 }
274 // ===== HTTP ERROR =====
275 else {
276     lcd.clear();
277     lcd.print("HTTP Error");
278     soundFail();
279 }
280 http.end();
281 delay(3000);
282 lcd.clear();
283 }
284 }
285 // =====
286 // ===== GET FINGERPRINT =====
287 // =====
288 int getFingerprintID() {
289     uint8_t p = finger.getImage();
290     if (p != FINGERPRINT_OK)
291         return -1;
292     p = finger.image2Tz();
293     if (p != FINGERPRINT_OK)
294         return -1;
295     p = finger.fingerFastSearch();
296     // ===== SUCCESS =====
297     if (p == FINGERPRINT_OK) {
298         return finger.fingerID;
299     }
300     // ===== UNKNOWN FINGER =====
301     if (p == FINGERPRINT_NOTFOUND) {
```

```
sketch_may11a.ino

302     lcd.clear();
303     lcd.print("Unknown Finger");
304     soundFail();
305     delay(2000);
306     return -1;
307 }
308 return -1;
309 }
310 // =====
311 // ===== ENROLL FINGER =====
312 // =====
313 void enrollFinger(int id) {
314     int p = -1;
315     lcd.clear();
316     lcd.print("New User ID:");
317     lcd.print(id);
318     // ===== FIRST SCAN =====
319     while (p != FINGERPRINT_OK) {
320         p = finger.getImage();
321         lcd.setCursor(0, 1);
322         lcd.print("Place Finger");
323     }
324     soundAlert();
325     p = finger.image2Tz(1);
326     if (p != FINGERPRINT_OK)
327         return;
328     // ===== REMOVE FINGER =====
329     lcd.clear();
330     lcd.print("Remove Finger");
331     delay(1000);
332     p = 0;
333     while (p != FINGERPRINT_NOFINGER) {
334         p = finger.getImage();
335     }
336     // ===== SECOND SCAN =====
337     p = -1;
338     lcd.clear();
```

```
sketch_may11a.ino

340     lcd.print("Place Again");
341     while (p != FINGERPRINT_OK) {
342         p = finger.getImage();
343     }
344     soundAlert();
345     p = finger.image2Tz(2);
346     // ===== CREATE MODEL =====
347     if (finger.createModel() == FINGERPRINT_OK) {
348         finger.storeModel(id);
349         lcd.clear();
350         lcd.print("Saved Success");
351         soundSaveMelody();
352         // SEND TO GOOGLE SHEET
353         sendDataToGoogleSimple(id);
354     }
355     // ===== FAILED =====
356     else {
357         lcd.clear();
358         lcd.print("Enroll Failed");
359         soundFail();
360     }
361     delay(2000);
362 }
363 // =====
364 // ===== SEND TO GOOGLE =====
365 // =====
366 void sendDataToGoogleSimple(int id) {
367     if (WiFi.status() == WL_CONNECTED) {
368         HTTPClient http;
369         String url =
370             googleScriptURL +
371             "?action=login&id=" +
372             String(id);
373         http.begin(url);
374         http.setFollowRedirects(HTTPC_STRICT_FOLLOW_REDIRECTS);
375         http.GET();
376         http.end();
```



5. Photos and Screenshots

Project Images:

